Module 27 – Cardiac Diagnostics Case Study

Objective

In this module, you will apply **machine learning techniques** to build a **Cardiac Disease Prediction Model**. The goal is to analyze patient health data and predict the likelihood of heart disease based on various medical attributes.

Reference

Watch the **Module 27 video** carefully. The concepts and workflow explained in the video will guide you in completing this case study step by step.

Dataset

You can use the **Heart Disease Prediction dataset** available on [Kaggle](). Search for "**Heart Disease UCI**" or "**Cardiac Diagnosis Dataset**".

🧩 Task Instructions

1. Import and explore the dataset (understand features such as age, cholesterol, blood pressure, etc.).

2. Perform data cleaning and preprocessing:

- Handle missing values (if any)

- Encode categorical variables

- Normalize or standardize data if needed

3. Split the dataset into **training** and **testing** sets.

4. Implement a **classification algorithm** (e.g., **Logistic Regression**, **Decision Tree**, **Random Forest**, or **KNN**) to predict heart disease.

5. Evaluate the model using **accuracy**, **confusion matrix**, **precision**, **recall**, and **F1-score**.

6. Write a brief summary of your analysis and key findings.

💾 Submission Requirements

- Submit your `.ipynb` file (Jupyter Notebook).

- Upload your notebook to **Google Colab** and share the **Colab link**